

CSCE 753 COMPUTER VISION · MODULE 10 · LECTURE NOTES

Transformers and Self-Supervision

Learning without labels, and attention over patches.

MODULE OUTCOMES

Explain the vision transformer and how it differs from a CNN.

Explain what inductive bias buys and costs.

Explain the main self-supervised objectives.

Explain contrastive language-image pretraining and what it enables.

State what self-supervised features are and are not good for.

1 Vision transformers

A VISION TRANSFORMER IS CSCE 636'S TRANSFORMER WITH PATCHES AS TOKENS

Cut the image into non-overlapping 16×16 patches, flatten each into a vector, project it linearly, and add a positional embedding. The image is now a sequence of a few hundred tokens.

Then run a completely standard transformer encoder (CSCE 636 Module 06) **and classify from a prepended class token.** **There is nothing vision-specific in the body of the architecture at all** — which was the point of the paper and is why the result was striking.

Self-attention is global from the first layer: every patch attends to every other patch. **So there is no receptive field to grow or to design** (Module 08 §3), and the architectural failure mode of a too-small receptive field simply does not exist.

And the cost is quadratic in the number of patches, since attention compares every pair. **High resolution is therefore expensive in a way it is not for a convolutional network**, which is why windowed and hierarchical variants (Swin and its relatives) exist — they reintroduce locality for computational reasons, arriving back at something structurally close to a CNN.

	Convolutional network	Vision transformer
Inductive bias	Locality, translation equivariance, and a spatial hierarchy — all built in.	Almost none beyond the patch grid itself.
Small data	Substantially better — the built-in bias is correct for natural images and costs no data to acquire.	Substantially worse — it must learn from data what the CNN was given.

	Convolutional network	Vision transformer
Large data	Plateaus earlier.	Keeps improving, and eventually wins — the crossover is empirical and depends on the task.
Receptive field	Grows with depth; must be computed and designed for.	Global from layer one.
Cost in resolution	Linear in pixel count.	Quadratic in patch count.

INDUCTIVE BIAS IS A PRIOR, WITH THE SAME TRADE AS EVERY PRIOR

Convolution hard-codes two facts: nearby pixels are more related than distant ones, and a feature detector useful at one location is useful at another. Both are true of natural images, and both were established by Module 08 §1's observation that learned layer-1 filters converge on the same primitives regardless.

So a convolutional network begins with free, correct knowledge and needs correspondingly less data — which is exactly Module 01 §3's fourth strategy, built into the architecture rather than learned from examples.

And a correct prior becomes a ceiling once data is abundant, because the architecture cannot represent the exceptions to it. **Translation equivariance is simply wrong for an image with a consistent horizon**, where 'sky' at the top and 'sky' at the bottom mean different things, and a CNN cannot easily learn that distinction.

Which is the bias–variance trade of CSCE 633 Module 03 in architectural form. It is why the CNN-versus-transformer question has a data-dependent answer rather than a winner, and why the honest recommendation below roughly ten million images is a CNN or a hybrid — a claim about your dataset, not about the architectures.

2 Self-supervision

CONTRASTIVE (SimCLR, MoCo)

two augmentations of one image should embed close together; different images should embed far apart.
 NEEDS MANY NEGATIVES -- a large batch, or a queue of past embeddings.
 Risk: it learns exactly the invariances your augmentations encode, which is a feature when they match your task and a hard limitation when they do not. Colour-jitter augmentation teaches the model to ignore colour, which is wrong for bird species.

SELF-DISTILLATION (BYOL, DINO)

a student network predicts a teacher's output on a different view of the same image; the teacher is an exponential moving average of the student.
 NO NEGATIVES NEEDED, which was genuinely surprising, and the reason it does not collapse to a constant is still debated.
 DINO's attention maps segment objects without ever being told that objects exist.

MASKED PREDICTION (MAE, BEiT)

mask 75% of the patches and reconstruct them.
 Directly CSCE 638's masked language modelling, applied to patches. Trains cheaply, because the encoder only sees the visible 25%.

THE ASYMMETRY THAT MATTERS: masked models fine-tune excellently and LINEAR-PROBE POORLY; contrastive models produce features that are immediately linearly separable. Which you want depends entirely on your labelled budget.

That asymmetry is the practically actionable finding and it is rarely stated clearly. If you have very little labelled data and will train only a linear head (Module 08 §2), a contrastive or self-distilled model is the right choice. If you have enough to fine-tune, a masked model will usually end higher. Choosing by benchmark headline rather than by which protocol you will actually use is a common and expensive error.

3 Language supervision

CLIP: TRAIN IMAGE AND TEXT ENCODERS TO AGREE

Train an image encoder and a text encoder jointly so that a photograph and its caption embed close together, and mismatched pairs embed far apart — contrastive learning across two modalities, over hundreds of millions of image-text pairs scraped from the web.

Captions are supervision that already exists. No annotation campaign, no fixed label taxonomy, and **the label space is open rather than a closed list of classes** — which is the structural advance, not the architecture.

Which gives zero-shot classification directly: embed the candidate class names as text ('a photograph of a ...'), embed the image, and take the nearest. **No training on the target classes at all**, and it works well enough to be a serious baseline on tasks nobody collected data for.

And it gives a shared embedding space usable for retrieval and for conditioning generative models, which is why text-to-image systems are built on top of it. **The limitation is that it inherits the biases of web captions faithfully** — the associations in the training text become associations in the embedding space, which is Module 12 §4's subject and is not fixable by scaling.

4 What they are good for

- **Little labelled data and plenty of unlabelled data.** The core case, and the gains over supervised pretraining are large and reliable.
- **Retrieval and nearest-neighbour search. A good embedding is directly useful with no head and no fine-tuning** — which is CSCE 670's subject, and is where self-supervised features are least caveated.
- **Dense correspondence. DINO features match across object instances and across wide viewpoint changes surprisingly well**, giving semantic correspondence between different objects of the same category — which is **Module 03's problem solved sideways**, and the result in this module most worth knowing, because it connects the learned half of the course back to the geometric half.
- **And initialisation for essentially anything**, including components of the geometric pipelines of Modules 03 through 07.
- **Not for: metric geometry, calibrated uncertainty, or anything requiring a guarantee. Module 08 §4's limits are entirely unchanged by scale or by self-supervision**, and it is worth saying so explicitly because the breadth of these models invites the assumption otherwise.

WHAT SCALE DID AND DID NOT DELIVER

Scale delivered representation quality. Features from a large self-supervised model now beat supervised ImageNet features on most transfer tasks — which was not true in 2018 and was not widely expected.

It delivered open-vocabulary recognition through language supervision (§3), which no amount of fixed-class training achieves, because the limitation there was the taxonomy rather than the data volume.

It did not deliver robustness. Large models are more robust than small ones and still fail under distribution shift (Module 12), **and they fail confidently** — the calibration problem of Module 08 §4 is not improved by scale and in some respects is worsened by it.

And it did not deliver geometry. CLIP cannot count reliably, struggles with spatial relations ('the cup left of the book'), and has no metric scale whatsoever — because Module 01's theorem is a theorem. **The geometric half of this course is not obsolete, and that is the honest summary of where the two halves stand relative to each other.**

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Dosovitskiy et al. — An Image is Worth 16x16 Words (ViT) (free)	The §1 architecture , including the data-volume crossover measurements of the comparison table.
Caron et al. — Emerging Properties in Self-Supervised Vision Transformers (DINO) (free)	The §2 self-distillation method , and the attention maps that segment objects unsupervised.
He et al. — Masked Autoencoders Are Scalable Vision Learners (free)	The §2 masked objective , with the probe-versus-fine-tune asymmetry reported honestly.
Radford et al. — Learning Transferable Visual Models from Natural Language Supervision (CLIP) (free)	The §3 result , and the limitations section is unusually candid and worth reading.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Implement patch embedding and a small ViT** and train it on a small dataset.
- 2 **Train a comparable CNN on the same data** and report which wins at 1k, 10k, and 50k training images.
- 3 **Visualise ViT attention maps** at several layers and describe what they attend to.

- 4 **Implement SimCLR's contrastive loss** and train on unlabelled images.
- 5 **Vary the augmentations** and report how the learned invariances change on a downstream task that needs colour.
- 6 **Linear-probe a masked model and a contrastive model** on the same task and report the gap.
- 7 **Fine-tune both** and report whether the ordering reverses.
- 8 **Run CLIP zero-shot** on a classification task of your own and compare against a trained baseline.
- 9 **Test CLIP on counting and spatial relations** and report the failures.
- 10 **Use DINO features for dense correspondence** between two different objects of the same category, and compare against SIFT.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Describe a vision transformer and say what is vision-specific in it.
- 2 Compare CNN and ViT on five axes.
- 3 Why is inductive bias both an advantage and a ceiling?
- 4 Describe the three self-supervised families and what each needs.
- 5 State the probe-versus-fine-tune asymmetry and its consequence.
- 6 What was CLIP's structural advance, and what does it enable?
- 7 Give four good uses of self-supervised features and three bad ones.
- 8 What did scale deliver, and what did it not?